

EE/CS 172/130
Digital Logic and Design

Topic 4 – Boolean Algebra

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Simplification of Logic Expressions

- Consider the following two expressions:

$$F1 = \bar{A}B(B + \bar{B})(\overline{A + C})$$

$$F2 = \bar{A}B\bar{C}$$

- Both generate the same result, but $F1$ consumes more hardware than $F2$.
- This means that both expressions are equivalent by $F2$ is simpler.
- There are mathematical and graphical methods which can be used to simplify an expression.

Boolean Algebra

- Boolean algebra is a set of rules with digital variables to develop, manipulate and simplify logic expressions.

Axiom / Postulates

- Set of logical expressions that we accept without proof; upon which we can build a set of useful theorems/properties.
- In Boolean algebra, the outputs of the three basic gates are axioms.

$$0 \cdot 0 = 0$$

$$0 \cdot 1 = 0$$

$$1 \cdot 0 = 0$$

$$1 \cdot 1 = 1$$

AND

$$0 + 0 = 0$$

$$0 + 1 = 1$$

$$1 + 0 = 1$$

$$1 + 1 = 1$$

OR

$$\overline{0} = 1$$

$$\overline{1} = 0$$

NOT

Properties in Boolean Algebra

Commutative Property

- $AB = BA$
- $A + B = B + A$

Associative Property

- $(AB)C = A(BC)$
- $(A + B) + C = A + (B + C)$

Distributive Property

- $AB + AC = A(B + C)$

Idempotent Property

- $AA = A$
- $A + A = A$

Identity Property

- $A \cdot 1 = A$
- $A + 1 = 1$

Null Property

- $A \cdot 0 = 0$
- $A + 0 = A$

Negation Property

- $A\bar{A} = 0$
- $A + \bar{A} = 1$

Double Negation Property

- $\bar{\bar{A}} = A$

Examples

- $F = ABC + CAB + BAC$
 $= ABC + ABC + ABC$ [associative]
 $= ABC(1 + 1 + 1)$ [distributive]
 $= ABC(1)$ [idempotent/OR]
 $= ABC$

- $F = ABC + ABC\bar{C}$
 $= AB(C + \bar{C})$
 $= AB(1)$
 $= AB$

- $F = AB(\bar{A}C + B)$
 $= AB\bar{A}C + ABB$
 $= 0 + AB$
 $= AB$

Practice Problems

- $F = \bar{X}Y + \bar{X}$
 $= \bar{X}$
- $F = X(\bar{X}YZ + \bar{X}Y\bar{Z})$
 $= 0$
- $F = (X + Y)(\bar{X} + XY)(\bar{Y} + X\bar{Y})$
 $= 0$
- $F = A\bar{B}\bar{C} + A\bar{B}C + \bar{A}BC + \bar{A}\bar{B}C$
 $= A\bar{B} + \bar{A}C$
- $F = (A + B)(\bar{A} + AB) + (\bar{B} + A\bar{B})$
 $= 1$
- $F = WXY(W\bar{X} + W\bar{Y}) + W\bar{X}Y(\bar{W}\bar{X} + X\bar{Y})$
 $= 0$
- $F = \bar{X}YZ + \bar{X}Y\bar{Z} + XZ$
 $= \bar{X}Y + XZ$
- $F = (WX + W\bar{Y})(X + W) + WX(\bar{X} + \bar{Y})$
 $= W(X + \bar{Y})$
- $F = (EP + \bar{E}N)(P\bar{E} + \bar{E}N + \bar{E}N + P\bar{N}) + (P\bar{E} + E\bar{N})(E\bar{N} + P\bar{N})$
 $= E\bar{N} + \bar{E}N + \bar{E}P\bar{N}$
- $F = (EP + \bar{E}N)(P\bar{E} + E\bar{N} + E\bar{N} + P\bar{N}) + (P\bar{E} + E\bar{N})(E\bar{N} + P\bar{N})$
 $= E\bar{N} + \bar{E}P$

The Absorption Property

- $A + AB = A$

Proof

$$\begin{aligned}\text{LHS} &= A + AB \\ &= A(1 + B) \\ &= A(1) \\ &= A = \text{RHS}\end{aligned}$$

- $A + \bar{A}B = A + B$

Proof

$$\begin{aligned}\text{LHS} &= A + \bar{A}B \\ &= A + AB + \bar{A}B \\ &= A + B(A + \bar{A}) \\ &= A + B(1) \\ &= A + B = \text{RHS}\end{aligned}$$

- $(A + B)(A + C) = A + BC$

Proof

$$\begin{aligned}\text{LHS} &= (A + B)(A + C) \\ &= AA + AC + AB + BC \\ &= A + AC + AB + BC \\ &= A(1 + C + B) + BC \\ &= A(1) + BC \\ &= A + BC = \text{RHS}\end{aligned}$$

Consensus Theorem (Included Factor Theorem)

- $AB + \bar{A}C + BC = AB + \bar{A}C$

Proof

$$\begin{aligned}\text{LHS} &= AB + \bar{A}C + BC \\ &= AB + \bar{A}C + BC(A + \bar{A}) \\ &= AB + \bar{A}C + ABC + \bar{A}BC \\ &= AB + ABC + \bar{A}C + \bar{A}BC \\ &= AB(1 + C) + \bar{A}C(1 + B) \\ &= AB(1) + \bar{A}C(1) \\ &= AB + \bar{A}C = \text{RHS}\end{aligned}$$

- $(A + B)(\bar{A} + C)(B + C) = (A + B)(\bar{A} + C)$

Proof

$$\begin{aligned}\text{LHS} &= (A + B)(\bar{A} + C)(B + C) \\ &= (A + B)(\bar{A} + C)(B + C + A\bar{A}) \\ &= (A + B)(\bar{A} + C)(B + C + A)(B + C + \bar{A}) \\ &= (A + B)(\bar{A} + C)(A + B + C)(\bar{A} + B + C) \\ &= (A + B)(A + B + C)(\bar{A} + C)(\bar{A} + B + C) \\ &= (A + B + 0C)(\bar{A} + C + 0B) \\ &= (A + B + 0)(\bar{A} + C + 0) \\ &= (A + B)(\bar{A} + C) = \text{RHS}\end{aligned}$$

Practice Problems

- $$\begin{aligned} F &= (A + C + D)(A + C + \bar{D})(A + \bar{C} + D)(A + \bar{B}) \\ &= (A + C + D\bar{D})(A + (\bar{C} + D)(\bar{B})) \\ &= (A + C + 0)(A + \bar{B}\bar{C} + \bar{B}D) \\ &= A + (C)(\bar{B}\bar{C} + \bar{B}D) \\ &= A + \bar{B}C\bar{C} + \bar{B}CD \\ &= A + 0 + \bar{B}CD \\ &= A + \bar{B}CD \end{aligned}$$

- $$\begin{aligned} F &= X\bar{Y} + X\bar{Y}Z \\ &= X\bar{Y} \end{aligned}$$

- $$\begin{aligned} F &= A\bar{B}C(AC + A\bar{B}C + AB) \\ &= A\bar{B}C \end{aligned}$$

- $$\begin{aligned} F &= X\bar{Y} + XYZ \\ &= X\bar{Y} + XZ \end{aligned}$$

- $$\begin{aligned} F &= \bar{W}X + Y + W\bar{Y}Z + \bar{W}\bar{X}\bar{Y}\bar{Z} + \bar{Y}\bar{Z} \\ &= X + \bar{Z} + W + Y \end{aligned}$$

- $$\begin{aligned} F &= \bar{A} + \bar{B} + ABC \\ &= \bar{A} + \bar{B} + C \end{aligned}$$

- $$\begin{aligned} F &= (A + \bar{A}\bar{B})(AC + A\bar{C}(B + \bar{B})) \\ &= A \end{aligned}$$

Prove:

- $$AB + \bar{A}C + BCD = AB + \bar{A}C$$

- $$\bar{X}\bar{Y} + \bar{Y}Z + XZ + XY + Y\bar{Z} = \bar{X}\bar{Y} + XZ + Y\bar{Z}$$

De Morgan's Theorem

- $\overline{AB} = \bar{A} + \bar{B}$
- $\overline{A + B} = \bar{A}\bar{B}$
- Both can be proved through truth tables.
- **Rule of thumb:** When dealing with multiple bars, start by analyzing the tallest bar first — it often simplifies the problem more quickly

Examples

- $$\begin{aligned} F &= \overline{A + \bar{B}CD} \\ &= \bar{A} \cdot \overline{\bar{B}CD} \\ &= \bar{A}(B + \bar{C} + \bar{D}) \\ &= \bar{A}B + \bar{A}\bar{C} + \bar{A}\bar{D} \end{aligned}$$
- $$\begin{aligned} F &= \overline{A + \overline{BC}} \\ &= \bar{A} \cdot \overline{\overline{BC}} \\ &= \bar{A}BC \end{aligned}$$

Practice Problems

- $F = \overline{(\bar{A}\bar{B})(\bar{C} + \bar{D})}$
 $= A + B + CD$

- $F = \overline{(X + \bar{Y}\bar{Z})(\bar{X}YZ)}$
 $= X + \bar{Y} + \bar{Z}$

- $F = \overline{\bar{A}(C + \bar{D})} + \overline{\bar{C}(A + \bar{B})}$
 $= A + \bar{C}(B + D)$

- $F = \overline{\bar{A}\bar{B}\bar{C} + \bar{A}B\bar{C} + A\bar{B}\bar{C}}$
 $= C + A\bar{B}$

- $F = \overline{\overline{A[B(\bar{C}\bar{D})]} \cdot \overline{(\bar{A}\bar{B}\bar{C})} B\bar{D}}$
 $= BC\bar{D} + A\bar{B} + A\bar{C}D$